WENKAI MA

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RESEARCH INTERESTS

Cross-Modal and Image-Text Retrieval · Multimodal Learning and Vision-Language Models · Fine-Grained Visual Recognition · Text-to-Image Person Re-Identification

Focus: Robust multimodal representation learning and cross-modal alignment for fine-grained retrieval and vision-language tasks, particularly under semantic ambiguity, visual similarity, and occlusion.

EDUCATION

M.Eng. in Electronic Information Sept. 2025 – June 2027 (Expected)

Anhui Agricultural University, Hefei, China

GPA: 4.40/5.00 | Cohort Rank: 1/80

Thesis: Image-Text Cross-Modal Retrieval Models for Goji Pest Recognition

Core Courses: Frontier Technologies of Electronic Communications (95/100); Intelligent Information Processing and Applications (94/100); Digital Image Processing and Analysis (91/100); Communication Theory and Systems (91/100); Modern Signal Processing Technology (90/100)

B.Eng. in Internet of Things Engineering Sept. 2021 – June 2025

City University of Hefei, Hefei, China

GPA: 3.91/5.00

Thesis: Design and Development of a Talent Cultivation Platform for Innovation and Entrepreneurship Research

Core Courses: Computer Networks, Data Structures, C/C++ Programming, Database Systems, Embedded Systems, Sensor Technology, Internet of Things Technology

PUBLICATIONS

Published

[1] **W. Ma** et al., "GjPest4CMR: LLMs-assisted morphology-aware fine-grained goji pest image-text retrieval," in *Proc. Int. Conf. Multimedia Retrieval (ICMR)*, 2026, pp. 138–146, doi: 10.1145/3805622.3810821.

Submitted

[2] **W. Ma** et al., "SaME-Align: Saliency-aware morphological semantic alignment for fine-grained goji pest image-text retrieval," submitted to *ACM Int. Conf. Web Search Data Mining (WSDM)*, 2027.

[3] **W. Ma** et al., "FAME-Align: Foreground-aware morphological evidence alignment for fine-grained goji pest image-text retrieval," submitted to *IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, 2027.

[4] **W. Ma** et al., "MIRAGE: Modality-aware identity-preserving relation alignment with graded evidence for text-to-image person re-identification," submitted to *ACM Web Conf. (WWW)*, 2027.

[5] K. Yu, W. Ma, **W. Ma**, J. He, T. Zhou, and Z. Cai, "UMS-Align: Uncertainty-guided semantic memory alignment for fine-grained point cloud-text matching," submitted to *ACM Web Conf. (WWW)*, 2027.

[6] K. Yu, W. Ma, **W. Ma**, J. He, T. Zhou, and Z. Cai, "Compatible or confusable? Consensus-guided hard-negative rectification for crop pest image-text retrieval," submitted to *ACM Web Conf. (WWW)*, 2027.

In Preparation

[7] **W. Ma** et al., "Beyond full visibility: Learning occlusion-robust text–image person retrieval from incomplete evidence," in preparation for *IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2027.

RESEARCH EXPERIENCE

Graduate Researcher June 2025 – Present

Key Laboratory of Agricultural Sensors, Ministry of Agriculture and Rural Affairs, Anhui Agricultural University, Hefei, China

Supervisor: Prof. Wentao Ma

- Independently conduct the full research cycle, including problem formulation, model development, experimental validation, and manuscript preparation.
- Supervise undergraduates through literature review, experimental design, implementation, and manuscript writing.

Fine-Grained Image-Text Retrieval for Goji Pest Recognition

- Constructed GjPest4CMR, a dataset of 9,496 images across 17 goji pest categories for fine-grained image-text retrieval, with LLM-assisted structured descriptions and expert verification.
- Proposed Structured Semantic Deconfounding for multi-granularity supervision and coarse-to-fine morphology learning, and Gated Adaptive Alignment for morphology-aware cross-modal alignment.
- Combined them in StruMorAlign, a parameter-efficient CLIP-based framework that achieved 78.72% mean recall on GjPest4CMR, 9.72 percentage points above a strong baseline [1].
- Developed two other alignment methods: SaME-Align, which is saliency-aware and deployed on edge devices [2], and FAME-Align, which is foreground-aware and needs no additional segmentation at inference [3].

Identity-Preserving Relation Alignment for Text-to-Image Person Re-Identification

- Investigated how conventional binary supervision in TIReID incorrectly treats semantically plausible non-ground-truth pedestrians as hard negatives.
- Proposed the concept of Cross-Identity Compatible Pairs to describe image-text pairs from different identities with highly compatible textual semantics.
- Developed MIRAGE, which models cross-identity compatibility from global and local graded evidence while preserving identity supervision, and calibrates image and text embedding distributions to narrow the modality gap. Validated on CUHK-PEDES, ICFG-PEDES, and RSTPReid [4].

Occlusion-Robust Text-Image Person Retrieval

- Developed a diffusion editing pipeline that inserts photorealistic foreground occluders while preserving pedestrian identity, clothing, pose, accessories, and scene consistency.
- Introduced identity and image-quality verification to filter out samples with identity drift, clothing alteration, malformed occluders, incorrect depth ordering, or excessive background modification.
- Built occlusion benchmarks for text-to-image person re-identification (TIReID) from CUHK-PEDES, ICFG-PEDES, and RSTPReid, preserving identity labels, textual descriptions, and official splits.
- Proposed MANTA, a retrieval framework that builds on an Identity-Relation Backbone, adding Masked Cross-Modal Alignment and Neighborhood Consistency Matching to make identity matching more stable and robust under occlusion [7].

FUNDED PROJECTS

Continual Fine-Tuning of Farmland Remote Sensing Large Models under Phenological Prior Constraints May 2026 – Present

National Natural Science Foundation of China, Young Scientists Fund (Category C)

Role: Core team member, responsible for figure preparation and core code design

- Aim: to address data distribution shifts in farmland remote sensing imagery across crop growth stages, regions, and time periods by integrating phenological priors into continual fine-tuning, while mitigating catastrophic forgetting.

Image-Text Cross-Modal Person Re-Identification in Open Scenarios June 2025 – Present

Anhui Provincial Natural Science Foundation, Youth Project (C)

Role: Core team member, responsible for figure preparation and core code design

- Aim: to develop fine-grained visual-textual representation learning and cross-modal semantic alignment methods for person re-identification under identity changes, background interference, occlusion, and image-text modality gaps.

HONORS & AWARDS

National Scholarship for Graduate Students (Ranked 1/80), Anhui Agricultural University Oct. 2026

Third-Class Academic Scholarship for Graduate Students, Anhui Agricultural University Oct. 2025

Excellent Volunteer (Ranked 1/63), Morita Therapy Academic Conference, Chinese Association for Mental Health Nov. 2024

SKILLS & LANGUAGES

Programming & Tools: Python, PyTorch, C/C++, MATLAB, Git, LaTeX, Linux, Photoshop

Deep Learning: CLIP, Vision Transformers, Contrastive Learning, Fine-Tuning

Computing: CUDA, GPU-Accelerated Training, Multi-GPU Experiment Management

Languages: Chinese (native), English (IELTS in preparation)

LEADERSHIP & SERVICE

President, Youth Media Center Dec. 2024 – Dec. 2025

City University of Hefei, Hefei, China

- Organized university events and handled photography, videography, and video editing.
- Operated the center's media account, which surpassed one million views.

Volunteer Nov. 2024

Morita Therapy Academic Conference, Chinese Association for Mental Health, Fuyang, China

- Responsible for event liaison and service.